Chapter 10

Cell Cycle and Cell Division

All organisms, even the largest, start their life from a single cell. Growth and reproduction are characteristics of cells, indeed of all living organisms.

One cell becomes millions

- all cells reproduce by dividing into two, each parental cell giving rise to two daughter cells every time it divides
- these newly formed daughter cells can themselves grow and divide, giving a new cell population formed by the growth and division of a single parental cell and its progeny
- such cycles of growth and division allow a single cell to form a structure consisting of millions of cells

What the cell cycle is

- during the division of a cell, DNA replication and cell growth also take place
- cell division, DNA replication and cell growth have to take place in a coordinated way, to ensure correct division and formation of progeny cells containing intact genomes
- the sequence of events by which a cell duplicates its genome, synthesises the other constituents of the cell and eventually divides into two daughter cells is termed the cell cycle

Growth is continuous, synthesis is not

- although cell growth, in terms of cytoplasmic increase, is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle
- the replicated chromosomes are then distributed to daughter nuclei by a complex series of events during cell division
- these events are themselves under genetic control

In this chapter

10.2 Phases of the cell cycle

10.4 M phase

10.5 Mitosis, stage by stage

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10.14 Significance of meiosis

Interphase, the long preparatory phase

M phase, where the actual division happens

One cell gives rise to two daughter cells

Figure 10.A A diagrammatic view of cell cycle indicating formation of two cells from one cell

10.1 Phases of the cell cycle

A typical eukaryotic cycle

- illustrated by human cells in culture, which divide once in approximately every 24 hours
- the duration can vary from organism to organism and from cell type to cell type
- yeast, for example, can progress through the cell cycle in only about 90 minutes

Interphase and M phase

- the M phase represents the phase when the actual cell division or mitosis occurs, and the interphase the phase between two successive M phases
- in the 24 hour average duration of a human cell cycle, cell division proper lasts only about an hour, the interphase lasting more than 95% of the duration
- the M phase starts with nuclear division, corresponding to the separation of daughter chromosomes, that is karyokinesis, and usually ends with division of cytoplasm, cytokinesis
- the interphase, though called the resting phase, is when the cell prepares for division by both cell growth and DNA replication, in an orderly manner

10.2 The three parts of interphase

G1 and S

- G1 phase, or Gap 1, corresponds to the interval between mitosis and initiation of DNA replication, during which the cell is metabolically active and continuously grows but does not replicate its DNA
- S or synthesis phase marks the period during which DNA synthesis or replication takes place, and the amount of DNA per cell doubles, so that if the initial amount is denoted 2C it increases to 4C
- chromosome number does not increase, a cell with 2n at G1 still having 2n after S phase

G2, and what happens where

- in animal cells, during the S phase, DNA replication begins in the nucleus and the centriole duplicates in the cytoplasm
- during the G2 phase, or Gap 2, proteins are synthesised in preparation for mitosis while cell growth continues

How do plants and animals continue to grow all their lives? Which tissues in higher plants have cells that divide all their life, and do animals have similar meristematic tissues?

10.3 The quiescent stage

Cells that stop dividing

- some cells in adult animals do not appear to exhibit division, such as heart cells, and many others divide only occasionally, as needed to replace cells lost through injury or cell death
- cells that do not divide further exit G1 phase to enter an inactive stage called the quiescent stage, G0, of the cell cycle
- cells in this stage remain metabolically active but no longer proliferate, unless called on to do so depending on the requirement of the organism

Where mitosis is seen

- in animals, mitotic cell division is only seen in the diploid somatic cells
- there are a few exceptions where haploid cells divide by mitosis, for example male honey bees
- against this, plants can show mitotic divisions in both haploid and diploid cells

10.4 M phase

The most dramatic period

- it involves a major reorganisation of virtually all components of the cell
- since the number of chromosomes in the parent and progeny cells is the same, it is also called equational division
- though for convenience mitosis has been divided into four stages of nuclear division, cell division is a progressive process and very clear-cut lines cannot be drawn between stages
- karyokinesis involves prophase, metaphase, anaphase and telophase

10.5 Prophase

What marks it

- prophase, the first stage of karyokinesis, follows the S and G2 phases of interphase, in which the new DNA molecules formed are not distinct but intertwined
- it is marked by the initiation of condensation of chromosomal material, which becomes untangled during chromatin condensation
- chromosomal material condenses to form compact mitotic chromosomes, seen to be composed of two chromatids attached together at the centromere
- the centrosome, having duplicated during S phase, begins to move towards opposite poles, each radiating out microtubules called asters, and the two asters with the spindle fibres form the mitotic apparatus

You have studied mitosis in onion root tip cells, which has 16 chromosomes in each cell. How many chromosomes will the cell have at G1 phase, after S phase, and after M phase? What will the DNA content be at G1, after S and at G2, if the content after M phase is 2C?

Chromatin condensing into compact chromosomes

Chromosomes at the equator, each chromatid joined by its kinetochore to a pole

Figure 10.B a and b A diagrammatic view of stages in mitosis

The end of prophase

- cells at the end of prophase, viewed under the microscope, do not show golgi complexes, endoplasmic reticulum, nucleolus and the nuclear envelope

10.6 Metaphase

Chromosomes at the equator

- the complete disintegration of the nuclear envelope marks the start of the second phase, so the chromosomes are spread through the cytoplasm
- condensation is completed and they can be observed clearly, so this is the stage at which the morphology of chromosomes is most easily studied
- a metaphase chromosome is made up of two sister chromatids held together by the centromere, and small disc-shaped structures at the surface of the centromeres, the kinetochores, serve as sites of attachment of spindle fibres
- all the chromosomes come to lie at the equator, one chromatid of each connected by its kinetochore to spindle fibres from one pole and its sister chromatid to fibres from the opposite pole

Key features

- spindle fibres attach to kinetochores of chromosomes
- chromosomes are moved to the spindle equator and aligned along the metaphase plate through spindle fibres to both poles
- the plane of alignment of the chromosomes at metaphase is referred to as the metaphase plate

10.7 Anaphase

Chromatids move apart

- at the onset, each chromosome arranged at the metaphase plate is split simultaneously, and the two daughter chromatids, now referred to as daughter chromosomes of the future daughter nuclei, begin their migration towards the two opposite poles
- as each chromosome moves away from the equatorial plate, its centromere remains directed towards the pole and so at the leading edge, with the arms trailing behind

Chromatids separating and moving to opposite poles

Chromatin collecting at each pole as two daughter nuclei form

The cytoplasm dividing, completing cell division

Figure 10.B c to e A diagrammatic view of stages in Mitosis

Key events of anaphase

- centromeres split and chromatids separate
- chromatids move to opposite poles

10.8 Telophase

The chromosomes decondense

- at the beginning of the final stage of karyokinesis the chromosomes that have reached their respective poles decondense and lose their individuality, and each set of chromatin material tends to collect at each of the two poles
- chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements
- a nuclear envelope develops around the chromosome clusters at each pole, forming two daughter nuclei
- nucleolus, golgi complex and ER reform

10.9 Cytokinesis

In animal cells

- mitosis accomplishes not only the segregation of duplicated chromosomes into daughter nuclei, but the cell itself is divided into two daughter cells by the separation of cytoplasm, called cytokinesis, at the end of which cell division is completed
- in an animal cell this is achieved by the appearance of a furrow in the plasma membrane, which gradually deepens and ultimately joins in the centre, dividing the cell cytoplasm into two

In plant cells

- plant cells are enclosed by a relatively inextensible cell wall, so they undergo cytokinesis by a different mechanism, wall formation starting in the centre of the cell and growing outward to meet the existing lateral walls
- the new cell wall begins with a simple precursor called the cell-plate, which represents the middle lamella between the walls of two adjacent cells
- at the time of cytoplasmic division, organelles such as mitochondria and plastids are distributed between the two daughter cells
- in some organisms karyokinesis is not followed by cytokinesis, so a multinucleate condition arises leading to a syncytium, as in the liquid endosperm in coconut

10.10 Significance of mitosis

What mitosis achieves

- mitosis, or equational division, is usually restricted to the diploid cells only, though in some lower plants and some social insects haploid cells also divide by mitosis
- it usually results in the production of diploid daughter cells with identical genetic complement, and the growth of multicellular organisms is due to mitosis
- cell growth disturbs the ratio between the nucleus and the cytoplasm, so it becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio
- a very significant contribution is cell repair, the cells of the upper layer of the epidermis, the lining of the gut and the blood cells being constantly replaced, while mitotic divisions in the meristematic tissues give continuous growth of plants throughout their life

10.11 Meiosis

Why the number must halve

- the production of offspring by sexual reproduction includes the fusion of two gametes, each with a complete haploid set of chromosomes, and gametes are formed from specialised diploid cells
- this specialised kind of cell division, which reduces the chromosome number by half and results in haploid daughter cells, is called meiosis
- meiosis ensures the production of the haploid phase in the life cycle of sexually reproducing organisms, whereas fertilisation restores the diploid phase
- we come across meiosis during gametogenesis in plants and animals, leading to the formation of haploid gametes

The key features

- meiosis involves two sequential cycles of nuclear and cell division, meiosis I and meiosis II, but only a single cycle of DNA replication
- meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S phase
- meiosis involves pairing of homologous chromosomes and recombination between non-sister chromatids of homologous chromosomes
- four haploid cells are formed at the end of meiosis II

Meiotic events group into meiosis I, with prophase I, metaphase I, anaphase I and telophase I, and meiosis II, with prophase II, metaphase II, anaphase II and telophase II.

10.12 Meiosis I

Prophase I, and its five stages

- prophase of the first meiotic division is typically longer and more complex than prophase of mitosis, and is subdivided into five phases based on chromosomal behaviour
- these are leptotene, zygotene, pachytene, diplotene and diakinesis
- during leptotene the chromosomes become gradually visible under the light microscope, and compaction continues throughout

Zygotene and pachytene

- during zygotene chromosomes start pairing together, and this process of association is called synapsis, such paired chromosomes being called homologous chromosomes
- electron micrographs indicate that synapsis is accompanied by formation of a complex structure called the synaptonemal complex, and the complex formed by a pair of synapsed homologous chromosomes is a bivalent or a tetrad
- during pachytene the four chromatids of each bivalent become distinct and clearly appear as tetrads, and the stage is characterised by the appearance of recombination nodules, the sites at which crossing over occurs between non-sister chromatids
- crossing over is the exchange of genetic material between two homologous chromosomes, an enzyme-mediated process involving recombinase, and recombination is completed by the end of pachytene, leaving the chromosomes linked at the sites of crossing over

Diplotene and diakinesis

- the beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes to separate from each other except at the sites of crossovers, these X-shaped structures being called chiasmata
- in oocytes of some vertebrates, diplotene can last for months or years
- the final stage is diakinesis, marked by terminalisation of chiasmata, during which the chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosomes for separation
- by the end of diakinesis the nucleolus disappears and the nuclear envelope breaks down, and diakinesis represents transition to metaphase

Metaphase I

- the bivalent chromosomes align on the equatorial plate
- the microtubules from the opposite poles of the spindle attach to the kinetochore of homologous chromosomes

Prophase I, with its five sub-stages

Metaphase I, bivalents on the equatorial plate

Anaphase I, homologous chromosomes separating

Telophase I, the dyad of cells

Figure 10.C Stages of Meiosis I

Anaphase I and telophase I

- in anaphase I the homologous chromosomes separate, while sister chromatids remain associated at their centromeres
- in telophase I the nuclear membrane and nucleolus reappear, cytokinesis follows, and this is called a dyad of cells
- although in many cases the chromosomes undergo some dispersion, they do not reach the extremely extended state of the interphase nucleus

Interkinesis

- the stage between the two meiotic divisions is called interkinesis, and is generally short lived
- there is no replication of DNA during interkinesis
- interkinesis is followed by prophase II, a much simpler prophase than prophase I

10.13 Meiosis II

Prophase II and metaphase II

- meiosis II is initiated immediately after cytokinesis, usually before the chromosomes have fully elongated, and in contrast to meiosis I it resembles a normal mitosis
- the nuclear membrane disappears by the end of prophase II and the chromosomes again become compact
- at metaphase II the chromosomes align at the equator and the microtubules from opposite poles of the spindle attach to the kinetochores of sister chromatids

Anaphase II

- it begins with the simultaneous splitting of the centromere of each chromosome, which was holding the sister chromatids together
- this allows them to move toward opposite poles of the cell, by shortening of the microtubules attached to kinetochores

Prophase II, chromosomes becoming compact again

Metaphase II, chromosomes at the equator

Anaphase II, sister chromatids separating

Telophase II, four haploid daughter cells

Figure 10.D Stages of Meiosis II

Telophase II

- meiosis ends with telophase II, in which the two groups of chromosomes once again get enclosed by a nuclear envelope
- cytokinesis follows, resulting in the formation of a tetrad of cells, that is four haploid daughter cells

10.14 Significance of meiosis

What meiosis conserves and what it varies

- meiosis is the mechanism by which conservation of the specific chromosome number of each species is achieved across generations in sexually reproducing organisms, even though the process itself, paradoxically, results in reduction of chromosome number by half
- it also increases the genetic variability in the population of organisms from one generation to the next
- variations are very important for the process of evolution

10.15 Summary

Where cell division fits

- according to the cell theory, cells arise from preexisting cells, and the process by which this occurs is called cell division
- any sexually reproducing organism starts its life cycle from a single-celled zygote
- cell division does not stop with the formation of the mature organism but continues throughout its life cycle

The cycle and mitosis

- the stages through which a cell passes from one division to the next is called the cell cycle, divided into interphase, a period of preparation for cell division, and mitosis, the actual period of cell division
- interphase subdivides into G1, when the cell grows and carries out normal metabolism and most organelle duplication occurs, S, which marks DNA replication and chromosome duplication, and G2, the period of cytoplasmic growth
- mitosis divides into prophase, metaphase, anaphase and telophase; chromosome condensation occurs during prophase while the centrioles move to opposite poles, the nuclear envelope and nucleolus disappear and spindle fibres start appearing
- metaphase is marked by alignment of chromosomes at the equatorial plate, during anaphase the centromeres divide and chromatids move towards the poles, and in telophase chromosomal elongation starts and nucleolus and nuclear membrane reappear, followed by cytokinesis

Meiosis

- in contrast to mitosis, meiosis occurs in the diploid cells destined to form gametes, and is called reduction division since it halves the chromosome number while making the gametes, the number being restored when two gametes fuse
- meiosis divides into meiosis I and meiosis II; in the first, homologous chromosomes pair to form bivalents and undergo crossing over, and its long prophase divides into leptotene, zygotene, pachytene, diplotene and diakinesis
- during metaphase I the bivalents arrange on the equatorial plate, followed by anaphase I in which homologous chromosomes move to opposite poles with both their chromatids, each pole receiving half the chromosome number of the parent cell
- in telophase I the nuclear membrane and nucleolus reappear; meiosis II is similar to mitosis, during anaphase II the sister chromatids separate, and four haploid cells are formed

10.16 Exercises

From this chapter

1 What is the average cell cycle span for a mammalian cell?
2 Distinguish cytokinesis from karyokinesis.
3 Describe the events taking place during interphase.
4 What is the quiescent phase of the cell cycle?

From this chapter, continued

1. Why is mitosis called equational division?
2. Name the stage of the cell cycle at which each of these occurs: (i) chromosomes are moved to spindle equator (ii) centromere splits and chromatids separate (iii) pairing between homologous chromosomes takes place (iv) crossing over between homologous chromosomes takes place
3. Describe the following, drawing a diagram to illustrate your answer: (a) synapsis (b) bivalent (c) chiasmata
4. How does cytokinesis in plant cells differ from that in animal cells?
5. Find examples where the four daughter cells from meiosis are equal in size and where they are found unequal in size.
6. Distinguish anaphase of mitosis from anaphase I of meiosis.
7. List the main differences between mitosis and meiosis.
8. What is the significance of meiosis?
9. Discuss with your teacher (i) haploid insects and lower plants where cell division occurs, and (ii) some haploid cells in higher plants where cell division does not occur.
10. Can there be mitosis without DNA replication in S phase?
11. Can there be DNA replication without cell division?
12. Analyse the events during every stage of the cell cycle and notice how these two parameters change: (i) number of chromosomes per cell (ii) amount of DNA content per cell